

CISC 667

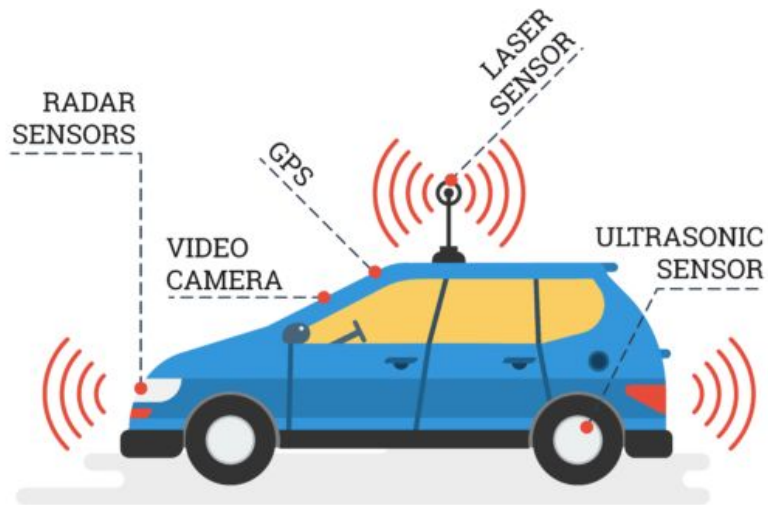
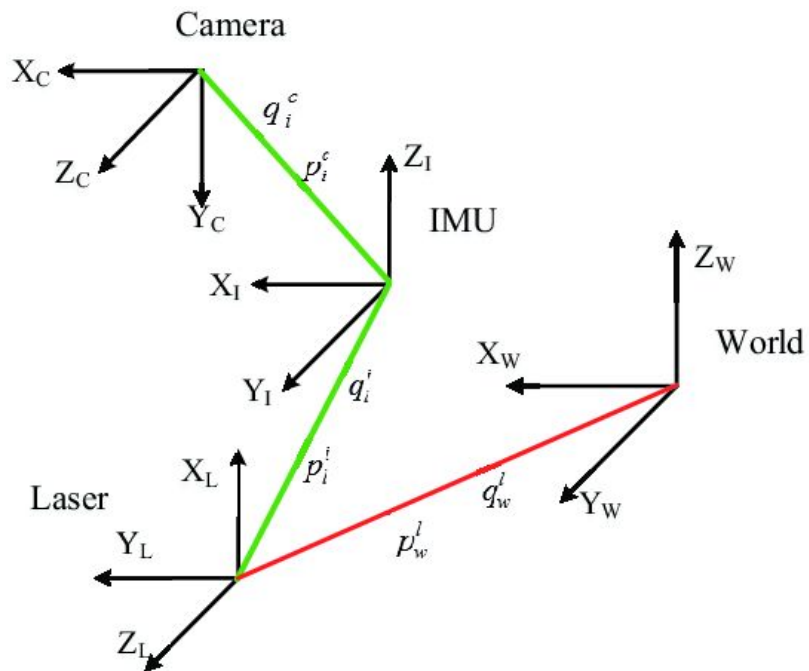
Introduction to Autonomous Driving

Lecture 5
Sept 17, 2025

Today's Outline

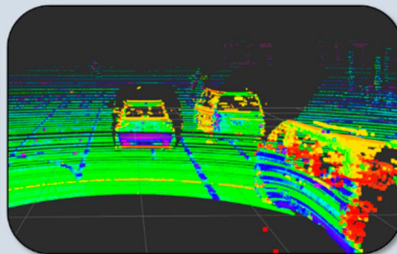
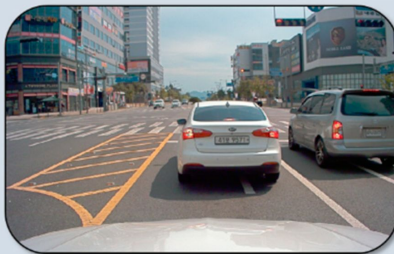
- Sensor Fusion.
- Project 1.

Sensor Frame Transformation

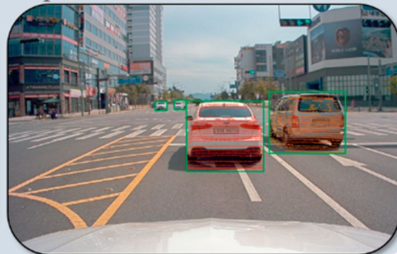
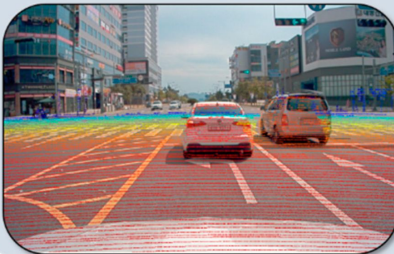


Sensor Fusion

Camera and LiDAR data



Data fusion and Depth estimation



Docker Container

- Check docker's help page
 - `sudo docker -h`
- List available docker images
 - `sudo docker image list`
- Check running docker instances
 - `sudo docker ps`
- Start a docker container
 - `sudo docker run -it <volume_name> /bin/bash`
- Attach to a running container
 - `sudo docker exec -it <volume_id> /bin/bash`

Example

Start camera

- `sudo -i`
- `ros2 launch realsense2_camera rs_launch.py enable_color:=true enable_depth:=true enable_rgbd:=true align_depth.enable:=true enable_sync:=true`

Start Yolo container

- `cd ~/NavOS/src/perception/util`
- `./run_container.sh`

Execute setup script

- `cd /perception_ws && ./src/util/setup.sh`

Compile package

- `colcon build --symlink-install`

Start Yolo script

- `source /opt/ros/humble/setup.sh && source install/setup.bash && ros2 run perception perception_node`

Visualize

- `sudo -i`
- `rviz2`